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Return to the Pre-Injury Level of Sport after Anterior Cruciate Ligament Reconstruction: A Practical Review with Medical Recommendations

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ABSTRACT

Returning to sport after anterior cruciate ligament reconstruction (ACLR) can be a challenging and complex process for the athlete, with the rate of return to the pre-injury level of sport observed to be less than athlete expectations. Of the athletes that do return to sport (RTS), knee re-injury rates remain high, and multiple studies have observed impaired athletic performance upon RTS after ACLR as well as reduced playing time, productivity, and career lengths. To mitigate re-injury and improve RTS outcomes, multiple RTS after ACLR consensus statements/clinical practice guidelines have recommended objective RTS testing criteria to be met prior to medical clearance for unrestricted sports participation. While the achievement of RTS testing criteria can improve RTS rates after ACLR, current criteria do not appear valid for predicting safe RTS. Therefore, there is a need to review the various factors related to the successful return to the pre-injury level of sport after ACLR, clarify the utility of objective performance testing and RTS criteria, further discuss safe RTS decision-making as well as present strategies to reduce the risk of ACL injury/re-injury upon RTS. This article provides a practical review of the current RTS after ACLR literature, as well as makes medical recommendations for rehabilitation and RTS decision-making after ACLR.

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Introduction

A complete tear of the anterior cruciate ligament (ACL) is a common knee injury [1], with an injury incidence-rate of approximately 85 per 100,000 within the 16 to 39 age group [1–3]. Regarding treatment options, anatomic ACL reconstruction (ACLR) is the gold standard treatment for managing complete ACL tears in young active patients who desire to return to jumping, cutting, and pivoting activities [4–8]. More specifically, ACLR with the bone-patellar tendon-bone autograft (BPTB) is the preferred surgical technique for athletes returning to high demand physical activities (i. e., Level 1 sport) [4, 5], because higher return to the pre-injury activity level and lower graft failure rates have been observed with the BPTB relative to the hamstrings tendon autografts (HT) [5, 6, 9, 10].

While the primary goal of ACLR is to restore knee stability and facilitate a safe return to high-demand physical activity, recent literature has revealed a discrepancy between patient expectations

of RTS and objective RTS outcomes after ACLR [11–13]. After being diagnosed with a complete ACL tear, most patients expect to return to their pre-injury activity level after undergoing ACLR [13]. After ACLR, an upwards of 90 % of patients may report normal or near normal knee function on the International Knee Documentation Committee (IKDC) Subjective Knee Form, but only 59–72 % of athletes may actually return to their pre-injury level of sport [11]. When discussing return-to-sport (RTS) outcomes after ACLR for the professional/competitive athlete, return to the pre-injury level of sport can vary more widely (i. e., 46–91 % of athletes returning to their pre-injury level of sport) [11, 14, 15], with some literature reporting suboptimal quality of performance in those that do RTS [16–20].

In an effort to improve RTS outcomes, multiple RTS after ACLR consensus statements/clinical practice guidelines have been published to help better define the RTS continuum, improve RTS rates, and reduce the risk of knee re-injury [12, 21–24]. Recently, the Panther Symposium ACL Injury Return to Sport Consensus Group defined RTS as the successful return to the pre-injury sports type, frequency, intensity, and quality of performance [12]. This definition of RTS necessitates the athlete to return to their pre-injury level of sport with adequate performance for successful RTS to be achieved, and with this, the emergence of performance based RTS testing and RTS criteria has become popular within clinical practice. However, while the achievement of RTS criteria during objective performance testing (i. e., a RTS testing battery) can improve RTS rates after ACLR [25–27], current RTS criteria does not appear valid for predicting safe RTS [28–30]. Of similar concern, a low RTS criteria achievement rate has been observed in athletes with and without a history of an ACL tear [31, 32], with only 17 % of athletes after ACLR and 24 % of non-injured athletes demonstrating symmetric performance compared to their contralateral limb (i. e., the achievement of 90 % between-limb symmetry) within a RTS testing battery [32].

Considering the discrepancies in the RTS after ACLR literature, there is a need to review the various factors related to the successful return to the pre-injury level of sport after ACLR, clarify the utility of objective performance testing and RTS criteria, further discuss safe RTS decision-making as well as present strategies to reduce the risk of ACL injury/re-injury upon RTS. Moreover, literature displaying the pragmatic implementation of current consensus statements and clinical practice guidelines is needed. Therefore, the purpose of this article is to provide a practical review of the published literature as it relates to return to the pre-injury level of sport after ACLR. More specifically, this article reports objective outcomes related to surgical technique, prehabilitation/rehabilitation, objective RTS testing, and psychological readiness for sport. This article also provides best-practice recommendations for facilitating the successful return of the athlete to their pre-injury level of sport after ACLR while mitigating the risk of ACL re-injury [12].

Search strategy

Google Scholar, Medline, and SPORTDiscus databases were searched from June 2023–November 2023 and included all study designs. As a practical review article, search strategies were similar to Gould et al. [33] but were not in accordance with the PRISMA guidelines or that of a scoping review [34, 35]. Studies published in the English language were included in this review article. Database searches were based around the terms ([ACL OR ACLR OR Anterior cruciate ligament reconstruction] AND [patellar tendon OR hamstrings OR autograft OR allograft] AND [prehabilitation OR rehabilitation OR injury prevention] AND [Sports OR RTS OR return-to-sport OR RTP OR return-to-play OR return-to-the-pre-injury OR return-to-performance OR performance] AND [RTP criteria OR RTP battery OR objective testing OR functional testing OR predictors OR psychometric OR re-injury]). Each author was assigned writing sections based on clinical expertise, and at least two authors were involved in independently screening titles and abstracts as well as reviewing identified articles in full text. References of included papers and consensus statements/clinical practice guidelines were also assessed for other relevant papers [12, 21–24, 33]. A total of 158 unique articles were identified, assessed, and utilized within this practical review.

The impact of the type of sport, sports-position and sex on RTS after ACLR

Cohort studies reporting outcomes after ACLR have observed a difference in RTS success based on sport, sport position, and gender (▶ **Table 1**). Sports that do not require land-based cutting/pivoting activities (i. e., Level-2 sport or below) appear to have a high return to the pre-injury level of sport, with 97 % of National Hockey League (NHL) athletes returning to professional hockey with no significant decrease in performance-based outcomes (▶ **Table 1**) [36]. Similarly, Erickson et al. [37] reported 80 % of Major League Baseball (MLB) athletes were able to return to their pre-injury level of sport after ACLR with no decrease in performance, regardless of ACL graft type or the performance of matched controls.

Athletes returning to Level 1 sports after ACLR, such as professional athletes within the National Basketball Association (NBA), Women's National Basketball Association (WNBA), National Football League (NFL), Major League Soccer (MLS), and the National Women's Soccer League (NWSL) have all been observed to experience a decrease in athletic performance upon their return to their pre-injury level of sport (▶ **Table 1**) [16–20, 38, 39]. In professional basketball, 86 % of NBA and 98 % of other professional-league athletes have been reported to return to their pre-injury level of sport [17, 20]; however, decreases in games/season, minutes/game, points/game, rebounds/game, and field goal percentages were observed [17, 20]. Linemen and skill position athletes (e. g., running backs and wide receivers) within the NFL appear to return to their pre-injury level of sport at a much lower rate than professional basketball athletes (linemen, 64.3 %; skill positions, 62 %) [18, 38]. Specifically, ACLR appears to significantly affect the position-specific performance of NFL wide receivers, with lower performance-based outcomes reported across a variety of metrics (▶ **Table 1**) [40]. Contrariwise, the quarterback position appears least affected by ACLR, with Erickson et al. [41] reporting 92 % of NFL quarterbacks returning to their pre-injury level of sport with no significant decrease in performance-based outcomes (▶ **Table 1**).

Sex-based differences in return to the pre-injury level of sport after ACLR also appear to exist. Compared to male soccer athletes within the MLS, female soccer athletes within the NWSL have been observed to RTS at a higher rate, with RTS rates of 77 % and 90 %, respectively [16, 19]. However, female athletes may experience a

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▶ **Table 1** Overview of studies reporting quality of performance outcomes after anterior cruciate ligament injury and/or reconstruction.

Study	ACLR Cohort	Study Period, Surgical Details, and Inclusion/ Exclusion	Criteria for RTS	Reported Performance-Based Outcomes	Significant Findings
Abed et al. [19]	Professional soccer athletes within the NWSL	Athletes who experienced an ACL tear and underwent surgery between 2013 and 2020 were eligible for inclusion. No surgical details provided or considered in selection.	Successful RTS was defined as any athlete who played at least 1 minute in at least 1 NWSL game after the injury.	RTS, games started, percentage of minutes played, +/- net per 90 minutes, goals scored, assists	RTS in the NWSL was 90 %. ▪ There was a significant decrease in quality of performance for minutes played, assists for forwards and midfielders, and goals scored.
Cinque et al. [18]	Professional football linemen within the NFL	NFL offensive and defensive linemen who sustained an ACL tear and underwent ACLR between 1980 and 2015 were eligible for inclusion. Graft type not specified. Athletes who suffered a meniscal tear, cartilage injury, MCL injury, or FCL injury (but not both at the same time) were included in the study. Athletes who suffered multiple ACL tears were included only if they had sufficient data for their first ACL tear. Multi-ligament knee injuries were excluded.	Successful RTS was defined as if an athlete played in at least 1 NFL game after their ACLR.	Mean time to RTS, number of pro bowls, seasons, games played, games started, games/ season within first through third season, games started within the first through third seasons defensive statistics, offensive statistics	ACL tears significantly hindered linemen within the NFL. ▪ 64.3 % RTS in the NFL ▪ Of those linemen who did return, there was no significant decrease in quality of performance.
Erickson et al. [36]	Professional hockey athletes within the NHL (Athletes <12-month follow-up were excluded)	NHL athletes who tore their ACL and underwent reconstructive surgery between 1990 and 2013 were eligible for inclusion. Graft type not specified. Any concomitant injuries excluded.	Successful RTS was defined as if an athlete played in any NHL game after ACLR.	Length of career, games played/ season, goals/season, assists/ season, points/season, +/- goal differential/season, penalty time/season, even-strength goals/season, short-handed goals/season, shots/season, shooting percentage/season, time-on-ice/season, time-on-ice/game	RTS in the NHL was 97 % with no significant difference in quality of performance. ▪ No decrease in career longevity, goals, and even-strength points/season. ▪ Observed improvement in mean shots and shooting percentage.
Erickson et al. [37]	Professional baseball athletes (MLB major or minor leagues)	All professional baseball athletes who underwent ACLR between 2010 and 2015 were eligible for inclusion. Grafts selection: ▪ Ipsilateral BPTB autograft (70%) ▪ Hamstring autograft (4-strand) (19%) ▪ Quadriceps autograft (1%) ▪ BPTB allograft (5%) ▪ Achilles allograft (4%) ▪ Hamstring allograft (1%) ▪ Tibialis anterior allograft (1 %) Surgical variables including graft type, femoral drilling technique, concomitant injuries, and others were recorded for each athlete.	Successful RTS was defined as returning to professional baseball and playing in at least 1 game.	Effect of graft type and femoral tunnel drilling technique on RTS, batting statistics, pitching statistics, level of play	RTS in the MLB was 80 % following ACLR. ▪ No difference between graft type or drilling techniques. ▪ No decrease in quality of performance and/or RTS rates versus matched controls.

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▶ **Table 1** Continued.

Study	ACLR Cohort	Study Period, Surgical Details, and Inclusion/Exclusion	Criteria for RTS	Reported Performance-Based Outcomes	Significant Findings
Erickson et al. [13]	Professional soccer athletes within the MLS	MLS athletes undergoing ACLR between 1996 and 2012 were eligible for inclusion. Graft type not specified. Certain concomitant knee injuries (when known) were deemed acceptable: ▪ Articular cartilage injury, meniscal tear, and MCL or FCL tear (but not both at same time). Exclusions: ▪ ACL + PCL and bicollateral ligament injuries were excluded. ▪ Athletes with < 8 months of follow-up were excluded.	Successful RTS was defined as if an athlete played in 1 MLS game following ACLR.	RTS rates, minutes/game, performance upon RTS, and difference versus controls	RTS in the MLS was 77%. ▪ Of those athletes who did return to the MLS after ACLR, no significant decrease in quality of performance was reported. ▪ No decrease in shots taken/season and assists versus controls.
Erickson et al. [41]	Professional football quarterbacks within the NFL	NFL quarterbacks with an ACL tear between 1988 and 2013 were eligible for inclusion. Graft type not specified. Certain concomitant knee injuries (when known) were deemed acceptable for inclusion. ▪ Articular cartilage injury, meniscal tear, and MCL or FCL injury (but not both at the same time) Those athletes who sustained a bicruciate (ACL and PCL) or combined ACL and bicollateral ligament injury were excluded.	Successful RTS was defined as if an athlete played in any NFL game after ACLR.	Games/season, completions/season, passing attempts/season, passing attempts/game, completion percentage/season, passing yards/season, passing yards/attempt, passing yards/game, passing touchdowns/season, interceptions/season, sacks/season, quarterback passer rating/season, fumbles/season	RTS for quarterbacks within the NFL was 92%. ▪ No significant difference in quality of performance. ▪ No significant difference in performance when matched against case-controlled athletes.
Harris et al. [17]	Professional basketball athletes within the NBA who sustained an ACL tear and ACLR between 1975 and 2012.	Athletes in the NBA who sustained an ACL tear and underwent ACLR between 1975 and 2012 were eligible for inclusion. Graft type not specified. Athletes with concomitant knee injuries (when known) were included: ▪ Articular cartilage injury, meniscal tear, and MCL or FCL tear (but not both at the same time).	Successful RTS was defined as if an athlete played in any NBA game after ACLR. Successful RTS in other leagues was defined as any game in either the NBA D-league or International Basketball Federation after ACLR.	RTS by league, career length, games/season, minutes, points, rebounds, assists, steals, blocks, annual salary, field goal percentage, free throw percentage, all-stars	High RTS rates were reported in any professional basketball league (98%), and RTS in the NBA was 86%. ▪ Of those returning to professional basketball, a decrease in quality of performance was observed in games/season, minutes, points, and rebounds/game along with field goal percentage. ▪ Except for games played, there was no significant difference in quality of performance against case controls.
Maman et al. [20]	Professional basketball athletes within the NBA	NBA athletes who suffered an ACL tear and underwent ACLR between 2012 and 2022 were eligible for inclusion. Graft type not specified. Exclusions ▪ Concurrent knee injuries (PCL/MCL/FCL) or other major leg injuries or Achilles tendon ruptures. ▪ Athletes with contralateral knee injuries	Successful RTS was defined as playing in the NBA following ACLR.	Minutes/game, points/game, rebounds/game, assists/game, steals/game, blocks/game, points/36-minutes, 2-point shooting percentage, 3-point shooting percentage, free throw shooting percentage	NBA athletes are greatly impacted by ACLR. ▪ Significant decrease in quality of performance was observed in minutes/game, points/game, rebounds/game, and assists/game.

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▶ **Table 1** Continued.

Study	ACLR Cohort	Study Period, Surgical Details, and Inclusion/Exclusion	Criteria for RTS	Reported Performance-Based Outcomes	Significant Findings
Manoharan et al. [40]	Professional football running back and wide receivers within the NFL	NFL running backs and wide receivers who underwent ACLR between 2009–2010 and 2015–2016 seasons were eligible for inclusion. No surgical details provided. Information concerning concomitant knee injuries was not readily available; as such, this was not recorded. Athletes with an ACL injury prior to the study period were excluded from the study group.	Successful RTS was defined by playing in at least 1 NFL game after ACLR.	RTS, games played, wide receiver yards/reception, receptions/game, receiving touchdowns/game, receiving touchdowns/season. Running backs – yards/rush, rushes/game, rushing touchdowns/season	ACL tear/ACLR resulted in an RTS rate of 61.8 % overall. ▪ Running backs returned to the NFL at 64.5 %. ▪ Wide receivers returned to the NFL at 60 %. ▪ For running backs who did return to the NFL, there was no difference in quality of performance in yards/rush, rushes/game, or rushing touchdowns/season in the ACLR group compared with a control group. ▪ For wide receivers who did return to the NFL, there was a significant decrease in quality of performance in receptions/game, yards/reception, and receiving touchdowns/season.
Mody et al. [38]	Professional football athletes within the NFL	NFL athletes who underwent ACLR between 2013 and 2018 were eligible for inclusion. No surgical data provided. No specific distinction was made for revision ACLR or multi-ligamentous injuries, and both were included in this study.	Successful RTS was defined as playing at least 1 snap in at least 1 regular season NFL game after ACLR.	Length of career post-injury, games played, games started, snap counts, approximate value (AV) algorithm to compare performance across positions	NFL athletes are severely hindered by ACL injury/ACLR. ▪ 28.5 % return to the league for more than 3 years. ▪ Running backs, defensive linemen, and linebackers had the lowest AV following injury. ▪ Quarterbacks were most likely to return and had superior performance compared to other positions.
Tramer et al. [39]	Professional basketball athletes within the WNBA	All ACL tears sustained in the WNBA from 1997 through 2018 were eligible for inclusion. Athletes were excluded if they sustained an ACL injury in the same knee before entering the WNBA or if they sustained documented concomitant injuries such as cartilage injuries and multi-ligamentous knee injury. Meniscal injuries not an exclusion factor.	Successful RTS was defined as any documented WNBA game play after ACLR.	RTS rates, total games played, total games started, minutes/game, points/game, field goals attempted, field goal percentage, attempted 3-points, 3-point percentage, attempted free throws, free throw percentage, rebounds/game, assists/game, steals/game, blocks/game	WNBA athletes were greatly affected by ACL tears/ACLR. ▪ 69.5 % were able to RTS after ACLR. ▪ Of those athletes able to RTS after ACLR, a significant decrease in quality of performance for the first year was reported, but this difference was not apparent after 3 years.

Description: The type of sport, sport position, and gender of an athlete can all significantly impact the rate of return to the pre-injury level of sport after anterior cruciate ligament reconstruction (ACLR) as well as performance-based outcomes upon return to sport (RTS). Abbreviations: ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; AV, approximate value algorithm; BPTB, bone-patellar tendon-bone; FCL, fibular collateral ligament; MCL, medial collateral ligament; MLB, Major League Baseball; MLS, Major League Soccer; NWSL, National Women's Soccer League; NFL, National Football League; NHL, National Hockey League; NBA; National Basketball Association; PCL, posterior cruciate ligament; WNBA, Women's National Basketball Association; RTS, return to sport.

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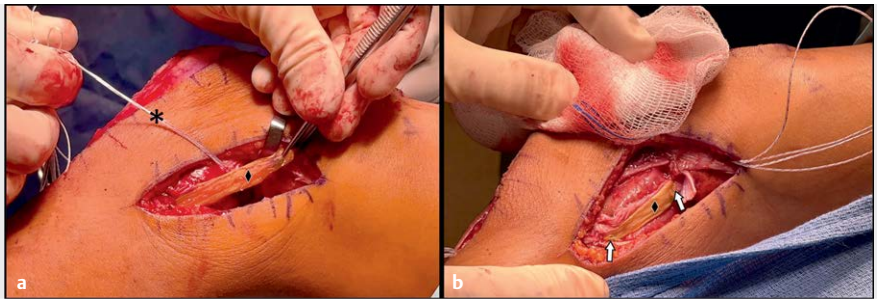

▶ **Fig. 1** Anterior cruciate ligament reconstruction (ACLR) with a lateral extra-articular tenodesis (LET). **Description:** The lateral extra-articular tenodesis (LET) procedure involves harvesting a graft from the inferior third of the iliotibial band (ITB) to reconstruct the anterolateral ligament of the knee (**a**). The addition of a LET at the time of ACLR can partially unload the anterior cruciate ligament (ACL) graft and lower the risk of graft failure as well as reduce anterolateral rotational laxity within the knee (**b**). Black asterisk: suture from the suture anchor used to fixate the femoral attachment of the LET 5 mm posterior and proximal to the FCL attachment on the femur. Black diamonds: the resected inferior third of the ITB used during the LET procedure. White arrows: suture anchor fixation of the proximal and distal LET at the anatomic attachments of the native anterolateral ligament.

greater impairment in athletic performance after ACLR than male athletes, as lower performance-based outcomes have been observed within the NWSL relative to the MLS (▶ **Table 1**) [16, 19]. Collectively, these observations emphasize the importance of athlete and sport-specific medical considerations after ACLR; the type of sport, sport position, and gender of an athlete can all significantly impact the rate of return to the pre-injury level of sport after ACLR as well as an athlete's quality of performance upon returning to sport.

ACLR technique and RTS

The elected surgical technique for ACLR appears to influence return to the pre-injury level of sport outcomes. Certain surgical factors, including graft choice for ACLR [42], augmentation of the ACLR to include a lateral extra-articular procedure (LEAP) [43], and concomitant meniscal repair/ligamentous reconstruction at the time of ACLR have all been observed to impact outcomes related to an athlete's return to their pre-injury level of sport [44]. Of these factors, graft selection appears particularly important, because higher RTS rates have been observed after ACLR with a BPTB autograft than with HT autografts or an all-soft tissue allograft [10, 42, 45]. Specifically, a cohort study by Gupta et al. [46] reported athletes had a significantly higher RTS rate after ACLR with a BPTB autograft relative to an HT autograft (74 % vs. 53 %, respectively). Preliminary data on the quadriceps tendon (QT) autograft for ACLR has shown promising results [47–51], but more robust studies investigating return to the pre-injury level of sport outcomes and graft re-injury rates in athletes are needed [47, 48].

More recently, biomechanical studies have suggested the addition of a lateral extra-articular tenodesis (LET) or anterolateral ligament (ALL) reconstruction (i. e., a LEAP procedure) at the time of ACLR can partially unload the ACL and lower the risk of graft re-injury as well as decrease anterolateral rotational laxity within the knee (▶ **Fig. 1**) [52, 53]. With this, evidence for the addition of a LEAP at the time of ACLR has emerged within cohorts at high risk of ACL graft failure [43, 52]. A cohort study of professional athletes by Hopper et al. [43] reported primary ACLR + LEAP had a 6.0 % failure rate at a minimum 2-year follow-up, compared to 15.5 % after an isolated ACLR. Likewise, Balendra et al. [15] reported the inclusion of an LET procedure at the time of ACLR with a BPTB autograft produced the lowest graft re-injury rates relative to other ACLR techniques (▶ **Fig. 1**), including an ACLR with a HT autograft with or without an LET.

Compared to an isolated ACLR, meniscal repair and/or other knee ligament reconstructions at the time of ACLR are known to reduce RTS rates [15, 44, 45, 54], and the number of reconstructed ligaments (i. e., 2 or more ligaments reconstructed at the time of ACLR) appears to influence RTS rates more than if the meniscus was intact versus repaired or resected [45, 54, 55]. A cohort study by Bakshi et al. [55] reported that 43.5 % of National Football League players were able to return to their pre-injury level of sport the next season after an ACL + medial collateral ligament (MCL) injury/surgery compared to only 18.5 % after an ACL, posterior cruciate ligament, and fibular collateral ligament injury/surgery. The impact of a 2-or-more ligament reconstruction at the time of ACLR may be even more detrimental in less elite cohorts, as Svantesson et al. [54] reported a 10 % return to the pre-injury level of sport after an ACLR + MCL reconstruction within the Swedish National Knee Ligament Registry compared to 48.9 % after an ACLR in isolation.

Within a cohort of professional soccer players, Balendra et al. [15] reported an RTS rate of 84 % after ACLR with a medial meniscus repair compared to 95.8 % after an isolated ACLR. Similarly, a subgroup analysis of secondary procedures after ACLR and prior to RTS reported undergoing a meniscal procedure delayed RTS by 4 months (14.0 versus 10.0 months) with 26 of the 222 athletes (11.7 %) requiring a subsequent meniscal surgery prior to RTS [15]. Regarding the impact of ACL graft selection on meniscal repair survivorship, Salem et al. [44] reported a 13.9 % meniscal repair failure rate after ACLR with a BPTB autograft at two- and six-year follow-up, compared to a respective 17.4 % and 19.4 % failure rate with an HT autograft or allograft. Although RTS rates were not reported, patients in the BPTB group appeared to be the youngest and still experienced the lowest rate of meniscal repair failure [44].

Collectively, literature investigating the impact of ACL graft selection on RTS suggests ACLR with the BPTB autograft may improve the rate of return to the pre-injury level of sport as well as increase ACL graft and/or meniscal repair survivorship [7, 44, 56]. For elite-level athletes or athletic cohorts at a high risk of ACL graft failure

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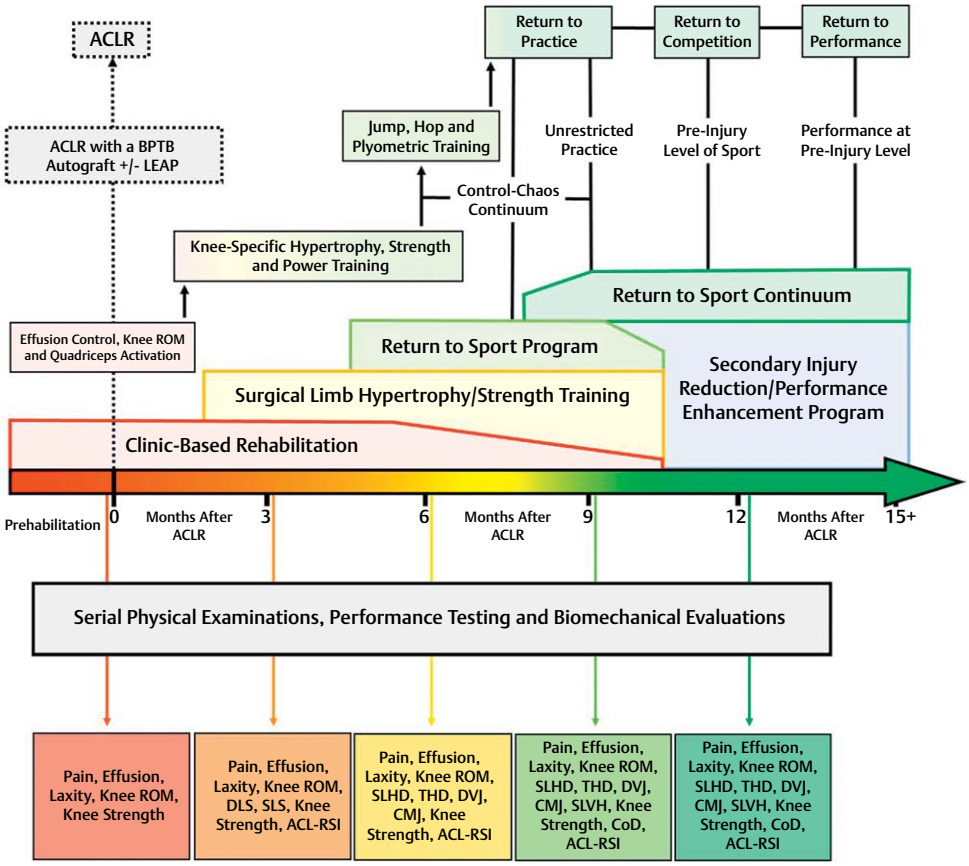

▶ **Fig. 2** A model of medical recommendations for return to the pre-injury level of sport after anterior cruciate ligament reconstruction (ACLR). **Description:** To optimize return to the pre-injury level of sports performance, prehabilitation should be completed prior to ACLR with a BPTB autograft. Postoperative rehabilitation should be progressed from clinic-based rehabilitation in isolation on to surgical limb hypertrophy/strength training as well a return-to-sport program. Serial physical examinations, performance testing, and biomechanical evaluations should be completed throughout the course of rehabilitation. Lastly, athletes should follow the return-to-sport continuum when progressing back to their pre-injury level of sport, and secondary injury reduction/performance enhancement programs may facilitate the return of on-field performance while mitigating the risk of secondary injury after ACLR. **Abbreviations:** ACLR, anterior cruciate ligament reconstruction; BPTB, bone-patellar tendon-bone; LEAP, lateral extra-articular procedure; ROM, range of motion; DLS, double-leg squat test; SLS, single-leg squat test; ACL-RSI, ACL-Return to Sport after Injury psychometric questionnaire; SLHD, single-leg hop-for-distance test; THD, triple hop-for-distance test; DVJ; drop-vertical jump maneuver/test; CMJ, counter-movement jump test; SLVH, single-leg vertical hop test; CoD, change-of-direction testing. Dashed lines: surgical procedure and time point. Black arrows: progression of exercise interventions. Colored arrows: example of time points for serial physical examinations, performance testing, and biomechanical evaluations. Black lines: progression within the return-to-sport continuum.

(e. g., a high school soccer athlete with genetic ligamentous laxity) [57], a concomitant LEAP procedure at the time of ACLR with a BPTB autograft may reduce the risk of ACL graft re-injury and improve return to the pre-injury level of sport outcomes (▶ **Fig. 2**) [15, 43, 52, 58].

Prehabilitation, rehabilitation and RTS training after ACLR

Prior to ACLR, it appears a prehabilitation program (i. e., preoperative rehabilitation program) is indicated to reduce joint pain and swelling within the involved knee, as well as improve knee-specific range of motion and strength (▶ **Fig. 2**) [59, 60]. Previous studies have investigated the impact of prehabilitation on long-term outcomes after ACLR [59, 61], for which Level 1 evidence has reported a positive association between prehabilitation and improved postoperative quadriceps strength [62–64], single leg hop test performance, subjective knee function, and RTS rates after ACLR [63, 64]. Although each cohort used a different rehabilitation protocol for preoperative and postoperative rehabilitation, a study by Failla et al. [60] reported 72 % of athletes receiving prehabilitation prior to ACLR returned to their pre-injury level of sport at a two-year follow-up compared to 63 % of those who received no prehabilitation. Similarly, Shaarani et al. [65] reported six weeks of prehabilitation reduced the mean RTS time from 42.5 weeks in athletes receiving no prehabilitation to 34.2 weeks in those completing prehabilitation.

After ACLR, compliance within a comprehensive rehabilitation program appears to improve the rate of return to the pre-injury level of sport. Han et al. [66] observed compliance within a super-

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vised rehabilitation program improved objective knee function as well as RTS rates after ACLR. Likewise, Della Villa et al. [67] observed compliance within a comprehensive rehabilitation program to be associated with improved RTS rates after revision ACLR, with 86 % of compliant athletes returning to their pre-injury level of sport [67]. Lastly, Edwards et al. [68] reported a positive association between the completion of a six-month, comprehensive rehabilitation program (i. e., the inclusion of jump, hop, and agility training during rehabilitation) and improved RTS rates/performance within a RTS testing battery; 37 % of the athletes who completed the rehabilitation program passed the study's RTS testing criteria compared to only 5 % of the athletes who underwent incomplete rehabilitation.

Regarding the evidence-based progression of an athlete back to their pre-injury level of sport after ACLR, the integration of RTS training programs within rehabilitation (i. e., advanced motor control, plyometric, and sport-specific training for the athlete) appears to be of high importance [12, 69–72]. Return-to-sport training programs can be implemented as an extension of standard-of-care rehabilitation after ACLR (▶ **Fig. 2**) [69–71, 73] and should focus on improving strength, motor control, movement quality, and sport-specific conditioning as well as help the athlete rebuild their training volumes back to a level that supports advancement within the RTS continuum (▶ **Fig. 2**) [12, 71, 73]. In their prospective cohort study designed to investigate the impact of an RTS training program on outcomes after ACLR, Nyland et al. [70] observed 84 % of their cohort returned to sport at or above their pre-injury level; a decrease in ipsilateral and contralateral knee re-injuries was also observed in those who completed RTS training as part of their rehabilitation program [70]. Similarly, a randomized control trial by Johnson et al. [69] observed an 87 % rate of return to the pre-injury level of sport in athletes completing RTS training as part of their rehabilitation. Lastly, a study by Arundale et al. [72] observed that athletes who participated in an RTS training program (and passed RTS criteria) were able to return to their pre-injury level of sport at a higher rate than those who did not; 78 % of athletes at one year and 95 % of athletes at two years returned to their pre-injury level of sport [72]. Collectively, these findings suggest compliance within prehabilitation, postoperative rehabilitation, and an RTS training program after ACLR can significantly increase the rate of return to the pre-injury level of sport while also reducing the risk of knee re-injury [12, 60, 62–72].

Objective performance testing and RTS after ACLR

Recent RTS after ACLR consensus statements/clinical practice guidelines have advocated for performance/RTS testing criteria to be met prior to an athlete's return to their pre-injury level of sport [12, 22]. Although the optimal performance/RTS testing battery is unknown [22, 74], objective testing should involve the assessment of athlete-specific functional skills, movement quality, strength, range of motion, balance, and motor control (▶ **Fig. 2**) [12, 22]. Of these tests, the objective measurement of quadriceps strength and strength-symmetry appears crucial, as previous studies have observed associations between greater surgical limb quadriceps strength and improved knee biomechanics [27, 75, 76], function and hop performance [25], patient acceptable symptom-state [26], and the maintenance of sports participation 1–10 years after ACLR [27].

Other components of a performance testing battery may include jump/hop testing, for which both qualitative and quantitative measurements should be collected (▶ **Fig. 2**) [12]. Recently, the utility of hop testing as part of the RTS testing battery has been questioned [77, 78], as the objective measurement of hop distance and/or height does not assess or account for the known intra-limb and inter-limb compensation patterns known to exist in individuals after an ACLR [77–79]. However, greater hop-for-distance performance within the surgical limb has been consistently associated with an improved return to the pre-injury level of sport after ACLR [30, 80–84], for which the use of the single and triple hop-for-distance tests is recommended (▶ **Fig. 2**) [85]. More recently, a cohort study by Kotsifaki et al. [86] advocated for an objective measurement of single-leg vertical jump performance as part of an RTS testing battery, because the quantitative measure of vertical jump performance may better identify inter-limb performance asymmetries than traditional hop-for-distance tests after ACLR. However, symmetric inter-limb single-leg vertical jump performance does not mean symmetric biomechanical movement patterns are present during the task, suggesting more sophisticated biomechanical evaluations may be indicated at the time of RTS testing [79].

Current RTS testing batteries and criteria do not appear valid for predicting safe RTS after ACLR [28]; and although a cohort study by Grindem et al. [87] reported an increased risk of ACL re-injury in patients failing to achieve RTS criteria for quadriceps strength prior to RTS, far more studies have observed either no association or an increased risk of ACL re-injury in patients achieving some components of RTS criteria within a variety of RTS tests [29, 88–95]. Specifically, a recent scoping review of prospective studies by Straub and Powers [95] reported an increased risk of secondary ACL injury in patients achieving symmetric quadriceps strength prior to RTS; the finding of an increased risk of secondary ACL injury in patients achieving RTS criteria may be especially true in athletes who obtain high psychometric test scores prior to RTS or who return to sport within one year of their ACLR [90, 96–99].

Of the various RTS tests described to assess safe RTS after ACLR, recent literature has identified some biomechanical variables associated with secondary ACL injury [89, 100], with some literature suggesting RTS testing batteries should utilize a maximum of five variables/tests (e. g., strength/power, movement biomechanics, proprioception, kinesthesia, and balance) when synthesizing information regarding safe RTS after ACLR [101, 102]. In a small prospective cohort study, Paterno et al. [100] reported that during a drop vertical jump maneuver, biomechanical variables related to net hip rotation moment impulse, frontal plane knee range of motion during landing, asymmetries in sagittal plane knee moments at initial contact, and postural stability were collectively a strong predictor (i. e., sensitivity, 0.92; specificity, 0.88) of a contralateral or secondary ipsilateral ACL injury after primary ACLR. More recently, King et al. [89] completed a prospective cohort study to examine differences in strength, jump, and change-of-direction performance/biomechanics between male athletes who re-injured their reconstructed ACL and those with no re-injury. At a two-year follow-up, no significant difference in strength, jump or timed change-of-direction performance was observed between groups [89]. How-

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ever, greater biomechanical asymmetries of trunk side-flexion, pelvic tilt, pelvic drop, and distance from the center of mass to the knee/ankle in the frontal plane during unplanned change of direction was observed within those sustaining a second ACL injury [89].

Collectively, recent literature suggests higher performance-based test scores during RTS testing may improve the prognosis for an athlete to successfully return to their pre-injury level of sport [30, 81–83, 103]. However, current RTS criteria may not be valid for predicting safe RTS [28, 30, 95], and more sophisticated biomechanical testing of known risk factors for ACL re-injury may be needed to more specifically synthesize information regarding safe RTS after ACLR (▶ **Fig. 2**) [77–79, 86, 88, 89, 101, 102].

Psychological readiness and RTS after ACLR

Severe time-loss injuries can trigger the onset or exacerbation of psychological/mental health disorders (i. e., depression, anxiety, and/or fear of re-injury) [104–106], for which a recent systematic review by Nwachukwu et al. [107] observed 65 % of athletes not returning to sport after ACLR reported a psychological factor as their primary reason for not returning. With this, recommendations for the assessment and monitoring of psychological/mental status after ACLR has become increasingly popular within the sports medicine community [12, 24, 48, 61]. More specifically, recent consensus statements/clinical practice guidelines have advocated for RTS decision-making to include information related to psychological readiness for sports participation [12, 24, 61], for which the ACL–Return to Sport after Injury (ACL-RSI) scale has been observed to be a valid instrument for identifying psychological readiness to RTS after ACLR [12].

Recent literature has reported a positive association between a higher ACL-RSI score and successful return to the pre-injury level of sport. In a prospective cohort study, Ardern et al. [108] found that athletes with a higher ACL-RSI score at the preoperative and 4-month postoperative time point (mean ACL-RSI of 45.2 and 57.3, respectively) were more likely to return to their pre-injury level of sport. The findings of this study were subsequently validated by a larger cohort study, in which Sadeqi et al. [109] reported an ACL-RSI score ≥ 60 at six months after ACLR was predictive of return to the pre-injury level of sport at a two-year follow-up. More recently, a systematic review by Xiao et al. [110] reported athletes who returned to sport after an ACLR had an average ACL-RSI score 20 points higher than those who did not return to sport; these findings support the recommendations put forth by recent consensus statements/clinical practice guidelines regarding psychological/mental assessment and monitoring [12, 24, 61], for which interventions implemented to optimize psychological readiness for sport after ACLR may improve outcomes related to return to the pre-injury level of sport [111].

While greater psychological readiness may improve the rate of return to the pre-injury level of sport, the relationship between psychological readiness and re-injury after ACLR appears more complex. Although low ACL-RSI scores have been associated with re-injury after ACLR [112], greater psychological readiness and knee-related self-efficacy has also been associated with an increased risk of ACL re-injury [29, 97, 98, 113]. The exact reason for an increase in ACL re-injury risk when high psychological readiness for sport is observed after ACLR is likely multifactorial in nature; however, higher knee-related confidence and self-efficacy when returning to sport may yield more exposure to intense/competitive sports participation within the first 12–24 months after ACLR [114, 115], increasing the risk of early sports-related ACL re-injuries after ACLR [29, 97, 98, 113]. Collectively, an inverted U pattern may best explain the relationship between psychological readiness and re-injury [116], for which too low or too high of an ACL-RSI score may increase an athlete's risk of re-injury after ACLR (**[Fig. 3]**). Whereas moderate to high ACL-RSI scores (e. g., ACL-RSI scores between 58–80) may improve outcomes related to return to the pre-injury level of sport while concurrently reducing the risk of ACL re-injury (**[Fig. 3]**) [97, 98, 108, 109, 112, 113].

Medical recommendations for rehabilitation after ACLR

Although anatomic ACLR may be required to restore knee stability to a level that supports a safe return to high-demand athletic activity [117, 118], the quality of an athlete's preoperative and postoperative rehabilitation program is of equal importance [12, 60, 62–72, 119]. Successful RTS after ACLR necessitates the postoperative recovery of knee range of motion [120], enough physical capacity to meet the demands of specific sporting task(s) [30, 80–83], the psychological desire/confidence/readiness for RTS [107–110] as well as the motor control strategies required for sport. With this, healthcare providers should provide both surgical and rehabilitation-based athlete education prior to the shared decision to undergo ACLR [12, 118].

Prior to ACLR, athletes should complete a prehabiltation program that emphasizes pain reduction and effusion control, the recovery of knee-specific range of motion, the optimization of quadriceps strength as well as thorough patient/athlete education [12, 59–61, 121]. The medical team (i. e., surgeon and rehabilitation specialists) should agree on the objective readiness of the knee for surgery, as preoperative knee range of motion and strength are significant predictors of the postoperative outcome after ACLR [63, 64, 119, 122, 123]; and although the reasons for surgical complications after ACLR are multifactorial [121, 124], a minimum of a two- to six-week period of prehabilitation (between ACL injury and ACLR) is indicated to reduce the risk of postoperative knee arthrofibrosis [121, 124]. Likewise, compliance within a six-week prehabilitation program may also accelerate postoperative recovery and the RTS timeline for an athlete [65].

When providing operative treatment to young/active athletes with ACL injuries, surgeons should consider performing ACLR with a BPTB autograft, because this graft can reliably facilitate the safe return of an athlete to their pre-injury level of sport (▶ **Fig. 2**) [10, 42, 45, 46, 100]. As previously mentioned, a LEAP is indicated at the time of ACLR for elite athletes and/or athletes with known non-modifiable risk factors for ACL graft laxity/re-injury (e. g., knee hypermobility, allograft, hamstring autograft, medial meniscectomy, and high tibial slope) [15, 43, 52, 57, 58, 125–128]. Postoperative rehabilitation should commence immediately after ACLR (i. e., within one to three days) with similar interventions as the aforementioned prehabilitation program (▶ **Fig. 2**) [129]. Specific postoperative interventions after ACLR should include the use of cryotherapy, neuromuscular electrostimulation, patellar mobilizations, and knee range of motion exercises [129]. The postoperative

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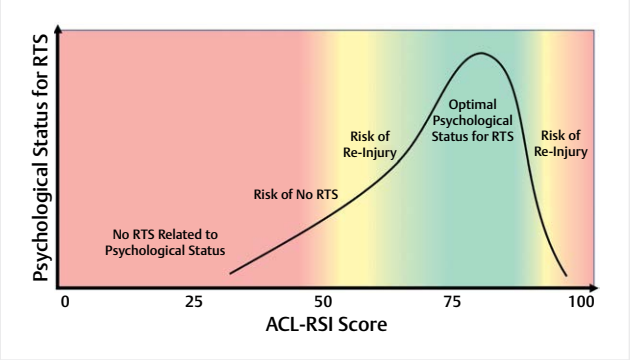

▶ **Fig. 3** A theoretical model of an "inverted U" relationship between psychological readiness for return to sport after anterior cruciate ligament reconstruction and knee re-injury risk. **Description:** An inverted U pattern may best explain the relationship between psychological readiness for sport after anterior cruciate ligament reconstruction (ACLR) and re-injury, for which too low or too high of an ACL–Return to Sport after Injury (ACL-RSI) score may increase an athlete's risk of knee re-injury. Moderate to high ACL-RSI scores may facilitate an athlete's return to their pre-injury level of sport while also reducing the risk of knee re-injury (i. e., produce optimal psychological readiness for return to sport). **Abbreviations:** RTS, return to sport; ACL-RSI, ACL–Return to Sport after Injury psychometric questionnaire.

recovery of active knee extension is of high importance [130], as the slightest postoperative loss of knee extension can provoke quadriceps inhibition and impair knee arthrokinematics [130] as well as increase the risk of post-traumatic knee osteoarthritis after ACLR [131, 132]. Weightbearing should commence immediately after an isolated ACLR with/without a LEAP, and home/community-based rehabilitation programs should be prescribed in conjunction with formal rehabilitation sessions [129].

A hypertrophy/strength training program is an important component of rehabilitation after ACLR (▶ **Fig. 2**) [129]. Due to ACL graft harvest from the extensor mechanism, a quadriceps resistance training load progression should be a cornerstone of rehabilitation after ACLR with a BPTB or QT autograft [48, 129, 133–135]. The load-progression for the quadriceps should start with non-weightbearing exercises and progress onto both open and closed kinetic-chain exercise progressions [48, 129]. The rehabilitation specialist should consider graft healing and joint homeostasis as key variables when developing and/or progressing the quadriceps exercise program [48]. Similarly, postoperative load-management should include the frequent monitoring of the surgical knee for any signs of an increasing knee effusion as well as for extensor mechanism pain at the graft harvest site [47, 48, 71, 136].

When discussing comprehensive rehabilitation and RTS after ACLR, RTS training programs should be considered a critical component of RTS success [12, 69–72]. Return-to-sport training programs should be provided as an extension of standard-of-care rehabilitation for the athlete, and RTS training should include formal strength and conditioning sessions; load progressions for jumping, hopping, plyometric, and change-of-direction exercise as well as motor control exercises that are specific to the sport/tasks the athlete to which is returning (▶ **Fig. 2**). Late-phase rehabilitation for the athlete should follow the recommendations put forth by current RTS consensus statements, which advocate for the application of the RTS continuum and a period of on-field rehabilitation [12, 24]. On-field rehabilitation should follow the control-chaos continuum and facilitate graded exposure to sports participation (▶ **Fig. 2**) [12, 24, 73, 137]; all relevant stakeholders (e. g., athlete, guardian, surgeon, team physician, rehabilitation specialist) should agree with the return-to-practice and competition progressions as well as clearance for unrestricted sports participation [12, 48, 138]. Lastly, secondary injury reduction/performance enhancement programs should be implemented throughout the RTS continuum (▶ **Fig. 2**) [48], because ongoing motor control/strength training may facilitate the absolute recovery of pre-injury athletic performance while reducing an athlete's motor expression of known biomechanical risk factors for ACL injury/re-injury [48, 139–142].

Medical recommendations for RTS after ACLR

Despite attracting substantial research attention [12], the optimal method to return athletes to sport after ACLR remains unclear [28, 74, 95, 143, 144]. A plethora of factors contribute to the decision to RTS after ACLR, including (but not limited to) the nature and severity of concomitant injuries to the knee (e. g., cartilage and meniscal injuries) [12, 145–147]; the biological healing of the ACL graft [12]; knee joint motion and homeostasis; the psychological and functional status of the athlete [12, 21–23]; rehabilitation compliance; the quality of the ACLR rehabilitation program [67, 71]; and contextual factors related to the type of sport, timing of season, athlete position, and level of competition [12, 138, 148]. With this, recent literature has highlighted the importance of shared decision-making after ACLR [12, 138, 148–150], for which RTS decisions should involve multidisciplinary input from various stakeholders [12].

Serial physical examinations and objective performance testing should be utilized throughout rehabilitation to confirm the effectiveness of exercise interventions, adjust the exercise prescription(s), and inform RTS decision-making/medical clearance for RTS [12, 48, 71]. Physical examinations should include the assessment of knee homeostasis (pain and/or effusion), stability and range of motion [12]. Serial performance testing should begin when the athlete is deemed ready (i. e., usually between three to four months after ACLR) and should be completed every two to three months thereafter until the shared decision to return the athlete to unrestricted sports participation is made (▶ **Fig. 2**) [12, 48, 71, 99, 134, 151]. The ACL-RSI should also be completed at these time points [12, 48, 61], as the ongoing assessment and monitoring of psychological status after ACLR provides an opportunity for targeted interventions (as needed) and may improve outcomes related to psychological readiness for sports participation [61, 107–111].

The objective measurement of quadriceps strength and jump/hop performance are fundamental components of a performance/RTS testing battery [27, 75, 76, 152], as these performance tests appear most associated with RTS success (▶ **Fig. 2**) [25, 30, 81–83, 103, 152]. Validated hop tests should be administered (e. g., single and triple hop for distance, single leg vertical jump for height), and data collection should include both qualitative and quantitative measurements [12]. When accessible, biomechanical evaluation(s) may most specifically provide information regarding safe RTS [88, 89, 100, 101], because the biomechanical analysis of

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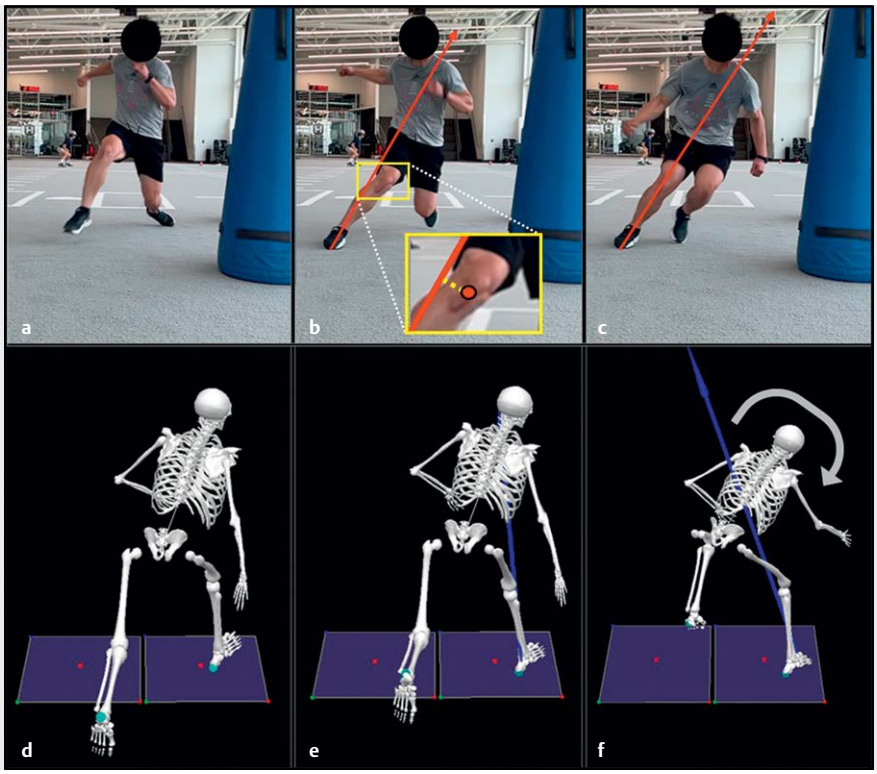

▶ **Fig. 4** Example of return-to-sport (RTS) testing of a change-of-direction (CoD) task. **Description:** Prior to commencing unrestricted sports participation, a final performance testing battery and/or biomechanical evaluation should be completed. Data collected during return-to-sport (RTS) testing can be utilized to further advise RTS decision-making as well as to implement and/or adjust ongoing injury reduction strategies. Sequence **a–c**: example of a 2-dimensional analysis of a change-of-direction (CoD) task utilizing a smartphone camera. Sequence **d–f**: example of a 3-dimensional biomechanical evaluation of a CoD task utilizing markerless motion-capture and force-plate technology. Red arrows: estimated ground reaction force-vector. Yellow callout box: visualization of knee abduction moment at initial ground contact. Dashed yellow line: depiction of external knee abduction moment. Red circle with black outline: approximate knee joint center. Blue arrows: ground reaction force-vector as collected during a biomechanical evaluation. Grey rotational arrow: presence of ipsilateral trunk side-flexion during a CoD task.

jump/hop/change-of-direction tasks may best identify known risk factors for ACL re-injury (▶ **Fig. 4**) [88, 89, 100, 101].

Of the various data synthesized for safe RTS decision-making after ACLR, some information appears more important than others [48]. The postoperative time frame between ACLR and RTS has been observed to be a modifiable risk factor for knee re-injury in non-elite/non-professional athletes [87, 153, 154], with current literature suggesting most athletes should wait a minimum of eight to nine months before returning to unrestricted Level 1 sports participation [12, 48, 87, 152–155]. More recently, risk calculator algorithms developed to estimate the risk of a secondary meniscal procedure and/or revision ACLR have been published for clinical use [127, 156–158]; these risk calculators use patient-specific data such as age, body mass index, preoperative knee laxity, activity level, and graft type to predict the risk of a secondary meniscal procedure and/or revision ACLR [127, 156–158]. Lastly, psychological readiness for sports participation appears highly associated with a successful return to the pre-injury level of sport [81, 107–110], for which the ACL-RSI should be used to assess an athlete's psychological readiness for sports participation after ACLR (**[Fig. 3]**) [12, 24, 61].

Prior to commencing unrestricted sports participation, a final physical examination, performance testing battery, and/or biomechanical evaluation should be completed (▶ **Fig. 2**) [12, 48]. All relevant information from the RTS testing battery should be clearly synthesized for RTS decision-making among the aforementioned stakeholders [138]. Special attention should be given to outcomes related to quadriceps strength, hop/jump test performance, and the ACL-RSI psychometric questionnaire, as these outcomes appear most associated with successful return to the pre-injury level of sport [30, 80–83, 108–110, 152]. When making medical recommendations regarding safe RTS after ACLR, healthcare providers should synthesize information related to non-modifiable risk factors for ACLR re-injury (e. g., knee hypermobility, allograft, hamstring autograft, medial meniscectomy, high tibial slope) [15, 43, 52, 57, 58, 125–128], for which published risk calculator algorithms can be used to estimate athlete-specific risk profiles [127, 156–158]. Lastly, if an athlete is returning to a sport with a high risk of knee re-injury, such as Level 1 sports, secondary injury reduction strategies should be implemented regardless of performance testing status or findings from a biomechanical evaluation (▶ **Fig. 2, 4**) [48, 159], because injury reduction programs appear highly effective at mitigating known biomechanical risk factors for

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ACL injury and can significantly reduce the overall knee re-injury incidence rate [139–142, 160].

Conclusion

While most athletes expect to return to their pre-injury level of sport after ACLR [13], recent literature has revealed a discrepancy between patient expectations of RTS and objective RTS outcomes after ACLR [11, 13–15]. Of the athletes that do RTS, high knee re-injury rates and suboptimal quality of performance has been observed [16–20]. Although multiple RTS after ACLR consensus statements/clinical practice guidelines have been published to improve RTS outcomes after ACLR [12, 21–24], additional medical recommendations may further optimize performance after ACLR and help facilitate an athlete's return to their pre-injury level of sport.

Prior to ACLR, athletes should complete a prehabilitation program that emphasizes effusion reduction and pain control within the involved knee, as well as interventions to improve knee range of motion and strength [59, 60]. Anatomic ACLR with a BPTB autograft with/without a LEAP may optimize an athlete's ability to return to their pre-injury level of sport while reducing their risk of knee re-injury [15, 43, 52, 58]. After ACLR, postoperative rehabilitation should place emphasis on athlete compliance within a comprehensive rehabilitation program, because comprehensive rehabilitation may improve RTS rates and reduce the risk of knee re-injury [12, 65–72].

Serial physical examinations, performance tests, and/or biomechanical evaluations should be completed throughout rehabilitation (▶ **Fig. 2**) [12, 99, 134, 151], with a final RTS testing battery administered prior to clearance for unrestricted sports participation [12, 48]; special attention should be given to outcomes related to quadriceps strength, hop test performance, and the ACL-RSI psychometric questionnaire, as these outcomes appear most associated with successful return to the pre-injury level of sport after ACLR [30, 80–83, 108–110, 152]. Information regarding safe RTS should be synthesized from known non-modifiable risk factors for ACL re-injury [161], published risk calculator algorithms, and/or the biomechanical evaluation of known risk factors for ACL injury/re-injury (▶ **Fig. 4**) [88, 89, 101, 127, 156–158]. Lastly, athletes should adhere to the RTS continuum when advancing from clinic to on-field rehabilitation, and secondary injury reduction/performance enhancement programs should be implemented regardless of athlete status at the time of RTS (▶ **Fig. 2**). These medical recommendations may improve athletic performance and reduce the risk of re-injury when returning athletes to sport after ACLR [139–142, 160].

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Conflict of Interest

The authors declare that they have no conflict of interest.

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